Digital Interaction Framework

Trust Architecture for Self-Sovereign Digital Relationships

5	6	4	7
Layers	Stages	Policies	ADRs

TRUST ARCHITECTURE

Layered Trust Model

05 Interaction `ixn`
Transactional vs Interactional. Shared context, activity binding, EWM. Trust investment scales with relationship duration.

04 Trust & Assurance `ACDC`
Fidelity, confidence, provenance. Progressive risk calibration from ephemeral to enduring relationships.

03 Technology `icp/rot`
KERI infrastructure, WebAuthn passkeys, ACDC credentials, AES-256-GCM, TLS 1.3, witness network, path history.

02 Governance `dip`
KEL governance (2-of-3), credential rules, organisational rules, Human Conductor oversight, delegation framework.

01 Policy Stacks `ADR`
Technical, Financial, Human, Agent domains. Most stable foundation layer. Rules, guidelines, conduct standards.

Layer Comparison

Layer	Actor	Primitive	Failure
05 Interaction	Both parties	ixn	Scope creep
04 Trust	Verifier	ACDC	Confidence loss
03 Technology	Infra	icp, rot	Key compromise
02 Governance	Conductor	dip	Authority gap
01 Policy	Foundation	ADR	Non-compliance

RELATIONSHIP LIFECYCLE

Discovery to Dissolution

01 Discovering `icp` `oobi`
AID publication, capability matching. Zero trust — all claims unverified until co-creation.

02 Co-Creating `ixn` `ACDC`
Bilateral credential exchange, context alignment, initial trust seeded through verification history.

03 Proposing `dip` `ACDC`
Terms encoded as ACDC schemas. Governance agreement. Agent delegation via dip if needed.

04 Using `ixn` `TEL`
Active operations. Continuous TEL verification. Trust accumulates. Anomalies escalate to conductor.

05 Updating `rot` `ACDC`
Key rotation (pre-committed). Credential refresh before expiry. Scope renegotiation anchored as events.

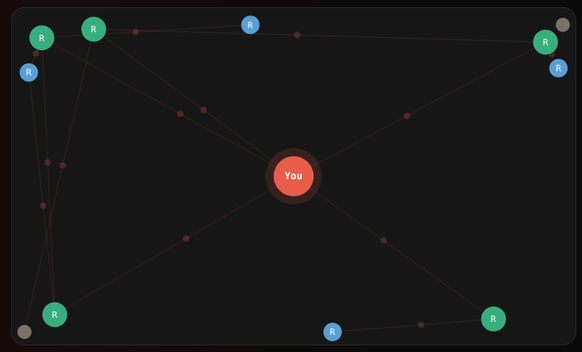
06 Archiving `TEL` `drt`
Credentials revoked. Agents deauthorised. Complete KEL/TEL history preserved permanently.

Trust by Stage

Stage	Trust	Credential	Reverse
Discovering	None	Outbound publish	Full
Co-Creating	Emerging	Bilateral	High
Proposing	Threshold	Terms codify	Moderate
Using	Established	Continuous	Low
Updating	Maintained	Refresh	Moderate
Archiving	Residual	Revocation	None

AGENT NETWORK

Trust Tier Architecture



● You ● Tier 1 ● Tier 2 ● Delegated

Tier 1 — Direct
Full ACDC chain, mutual KEL witness, signed channels, real-time TEL, delegation eligible.

Tier 2 — Shared
Group credential schema, issuer-mediated trust, attribute-based access, progressive elevation.

Teammate
Human Conductor + AI agent. Root AID dip delegated AID. 70/30 energy split. Escalation returns control.

Dependable
Witness (2-of-3), watcher, OOBI discovery, TEL registry. Geographic distribution required.

Protocol Infrastructure

`KEL` Key Event Log — icp, rot, ixn, dip, drt

`TEL` Transaction Event Log — credential lifecycle

`OOBI` Out-of-Band Introduction — discovery bootstrap

POLICY FRAMEWORK

Policy Stack Architecture

Technical
Ed25519, AES-256-GCM, TLS 1.3, HSM, 2-of-3 witness, OpenAPI 3.1, 7yr audit retention.

Financial
Tx limits, NPP/BPAY/SWIFT, AUSTRAC AML/CTF, APRA CPS 234, CDR consent, cross-border.

Human
Conductor 70/30, graduated disclosure, OAIC privacy, escalation chains, consent

Agent
Action whitelists, 24h time-limit, all ixn logged, anomaly escalation, ACDC-gated,

Architecture Decision Records

ADR-001	**KERI-First Identity** — All IDs are KERI AIDs. No alt primary.
ADR-002	**Passkey-Only AuthN** — FIDO2/WebAuthn. No passwords/SMS.
ADR-003	**ACDC Schema Gov.** — Foundation schemas. Graduated disclosure.
ADR-004	**Witness 2-of-3** — 3 witnesses, geo-distributed, 2-of-3 threshold.
ADR-005	**Agent via dip** — Scoped time-limited dip delegation.
ADR-006	**Pre-Rotation** — Next keys committed at inception. Mandatory.
ADR-007	**TEL for All** — Every ACDC has TEL. Check before accept.

Agent Autonomy Levels

Lvl	Name	Authority	Oversight
L1	Assistive	Read-only, suggest	Continuous
L2	Supervised	Execute w/ approval	Per-action
L3	Bounded	Within rules	Exception
L4	Autonomous	Self-directed	Periodic
L5	Strategic	Cross-domain	Outcome